

A disproof of conjecture 00000002131

Stanley–Wilf direct-sum multiplicativity

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Abstract

The one-element permutation has no avoiding permutations of positive size, so its Stanley–Wilf limit is zero. Its direct sum with itself is the increasing two-element permutation. At each size there is exactly one permutation avoiding that pattern, so its limit is one. Thus the displayed multiplicativity assertion would imply $1 = 0 \cdot 0$. We formalize the actual permutations, classical containment, avoidance cardinalities, direct sum, and real limits in Lean 4.

1 The source and the domain

Both language versions of conjecture 00000002131 assert

$$L(p \oplus q) = L(p)L(q),$$

and add that the exponential law of direct sums is a “complete characterization.” The displayed assertion is universal over finite patterns. The source imposes no lower bound on their lengths other than the ordinary nonempty-pattern convention. Our witness has length one in each input; it is an ordinary permutation, not an empty or repeated-value pattern. This length convention is essential to the stated counterexample and is not hidden.

The characterization clause specifies neither a characterized object nor a separate equivalence. We do not invent such a predicate. Refuting the displayed necessary assertion refutes the conjecture as written, regardless of any further clause that retains that assertion. No independent classification theorem about the qualitative extension is claimed.

2 Definitions and exact counts

Write $\mathfrak{S}_n$ for the bijections of $\{0, \dots, n-1\}$. A permutation $s \in \mathfrak{S}_n$ contains $p \in \mathfrak{S}_k$ when there is a strictly increasing map $e : \{0, \dots, k-1\} \rightarrow \{0, \dots, n-1\}$ satisfying, for every i, j ,

$$s(e(i)) < s(e(j)) \iff p(i) < p(j).$$

The positions need not be consecutive. Define $\text{Av}_n(p)$ to be the set of all $s \in \mathfrak{S}_n$ that do not contain p , and let $a_n(p) = |\text{Av}_n(p)|$. The direct sum of $p \in \mathfrak{S}_k$ and $q \in \mathfrak{S}_l$ is the bijection in $\mathfrak{S}_{k+l}$ given by

$$(p \oplus q)(i) = p(i) \ (i < k), \quad (p \oplus q)(k+j) = k + q(j) \ (j < l).$$

Let $u \in \mathfrak{S}_1$ and $v \in \mathfrak{S}_2$ be the identity permutations. These block equations give $u \oplus u = v$.

Lemma 1. *For every $n \geq 1$, $a_n(u) = 0$. For every $n \geq 0$, $a_n(v) = 1$.*

Proof. Every nonempty permutation contains u : select any one position. The required relative-order comparison on a one-element index set is vacuous. Thus no permutation of positive size avoids u .

A permutation contains v precisely when it has two positions $i < j$ with $s(i) < s(j)$. Since a permutation is injective, absence of such a pair is precisely strict decrease. There is exactly one strictly decreasing bijection of the finite ordered set, namely $i \mapsto n-1-i$. Indeed, strict decrease and bijectivity force the values in order to be $n-1, n-2, \dots, 0$; equivalently, there are exactly $n-1-i$ values below $s(i)$, so $s(i) = n-1-i$. This also covers $n=0$, where the unique empty bijection avoids v . Hence the full avoidance set is a singleton, and its cardinality is one. $\square$

3 Genuine limits and the contradiction

For each pattern, use the positive-index real root sequence

$$b_p(m) = (a_{m+1}(p))^{1/(m+1)} \quad (m \in \mathbb{N}),$$

where the count and exponent are real numbers. This is the usual sequence $a_n(p)^{1/n}$ along $n \geq 1$, reindexed by $n = m + 1$; no zeroth root or natural number division is involved. Write $\text{SW}(p, x)$ when b_p converges to $x \in \mathbb{R}$. It specifies a genuine real limit, not a limsup or a freely assigned growth value.

The lemma gives $b_u(m) = 0$ and $b_v(m) = 1$ for every m . The exponent is positive, so the real power of zero is zero. Consequently

$$\text{SW}(u, 0), \quad \text{SW}(v, 1), \quad L(u) = 0, \quad L(v) = 1.$$

Uniqueness of real limits identifies every possible limit value for these patterns with the displayed numbers.

Theorem 1. *The universal equality in conjecture 00000002131 is false.*

Proof. Substitute $p = q = u$. Since $u \oplus u = v$, the left side is $L(v) = 1$, whereas the right side is $L(u)L(u) = 0 \cdot 0 = 0$. $\square$

A fully specified version of the displayed universal assertion is: for all positive k, l and $p \in \mathfrak{S}_k, q \in \mathfrak{S}_l$, there exist real x, y, z such that

$$\text{SW}(p, x), \quad \text{SW}(q, y), \quad \text{SW}(p \oplus q, z), \quad z = xy.$$

The formal disproof negates this assertion using the actual limits above and their uniqueness. It does not rely on nonexistence of a limit, an arbitrary definition of L , or a table of unproved growth constants. Proving general Stanley–Wilf limit existence for unrelated patterns is unnecessary here. This counterexample is to the source as stated; it does not purport to settle an amended claim excluding one-element input patterns.

4 Formal verification and reproduction

The accompanying Lean project uses actual finite permutations, the classical containment predicate, and cardinalities of the full avoiding subtype. Its direct-sum construction and block formulas connect the permutation operation to the definition above. The count and convergence proofs are symbolic for all required sizes; no empirical enumeration or external numerical computation is needed.

The project pins Lean 4.19.0 and Mathlib v4.19.0. From the `lean/` directory, run:

```
lake build
lake env lean -DwarningAsError=true Conjecture2131.lean
lake env lean -DwarningAsError=true Check.lean
```

The accompanying verification records give the fresh build, full theorem-type and axiom inspection, exhaustive compiled-declaration inventory, exact source identities, and separate semantic review. Local validation is distinct from maintainer acceptance. The byte-identical bilingual conjecture is included as `conjecture.md`.